Wireframe Document

Supply Chain Management Analysis

Revision Number - 1.2

Last Date of Revision - 20/03/2025

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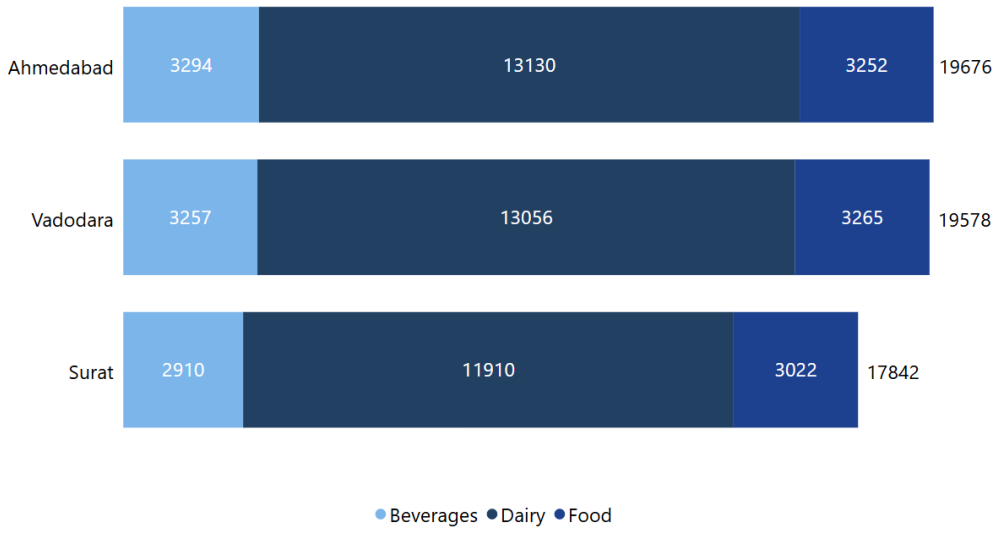
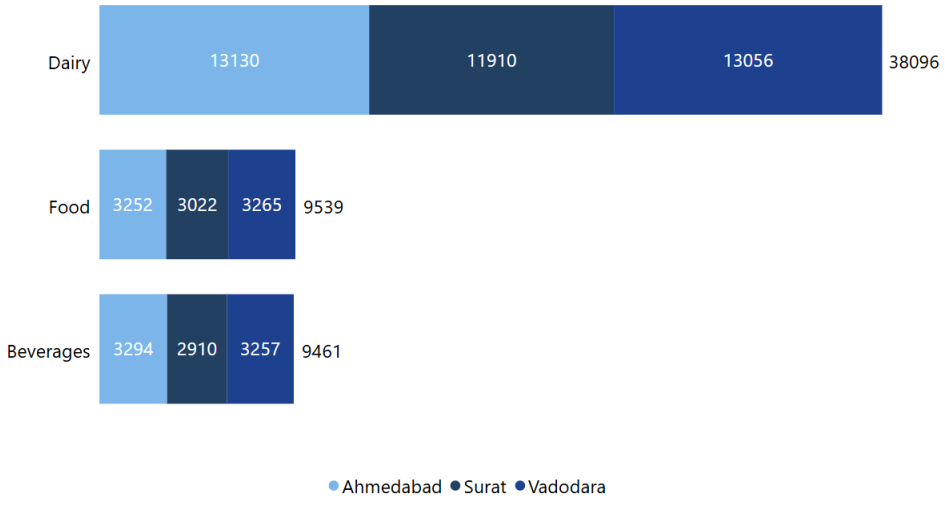
Document Control

Date	Version	Description	Author
10/03/2025	1.0	Introduction, Problem Statement	Shubham Tembhurne
10/03/2025	1.1	Dataset Information, Vi Architecture Description	Shubham Tembhurne
20/03/2025	1.2	Final Revision	Shubham Tembhurne

We Performed Exploratory Data Analysis on Quadratic AI and then created a Power BI Desktop Dashboard.

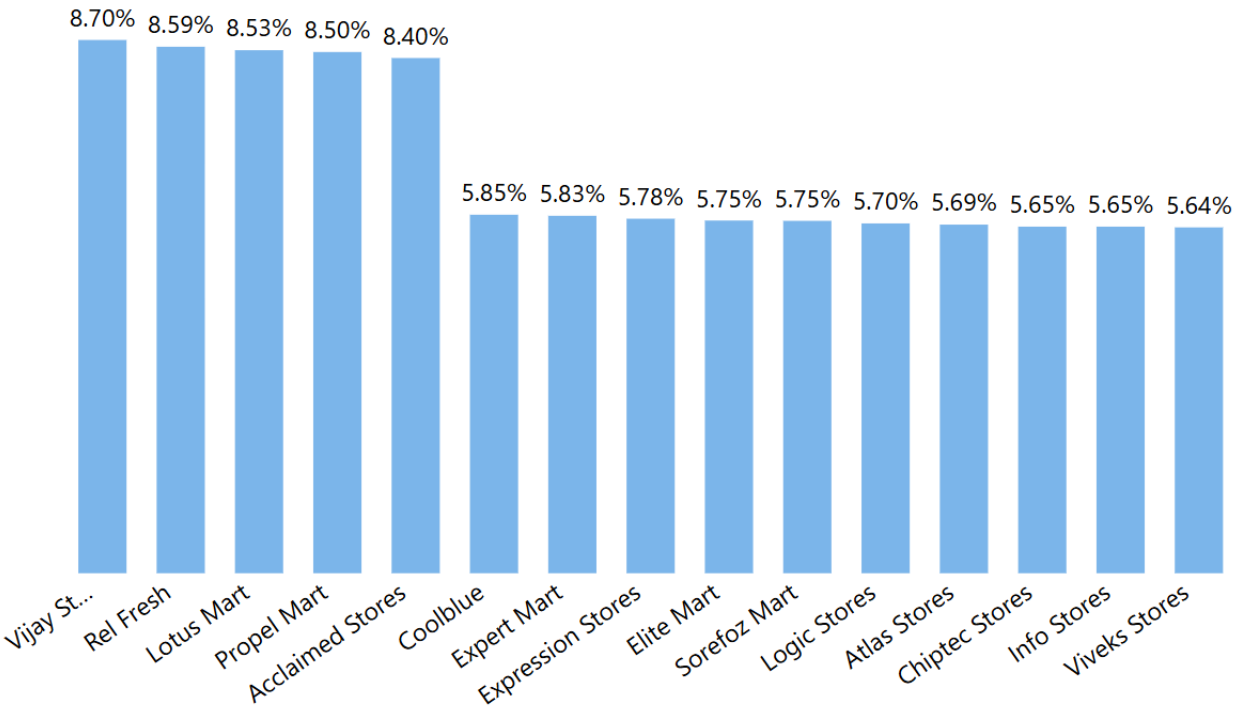
1. What is the city level performance and product at order level?





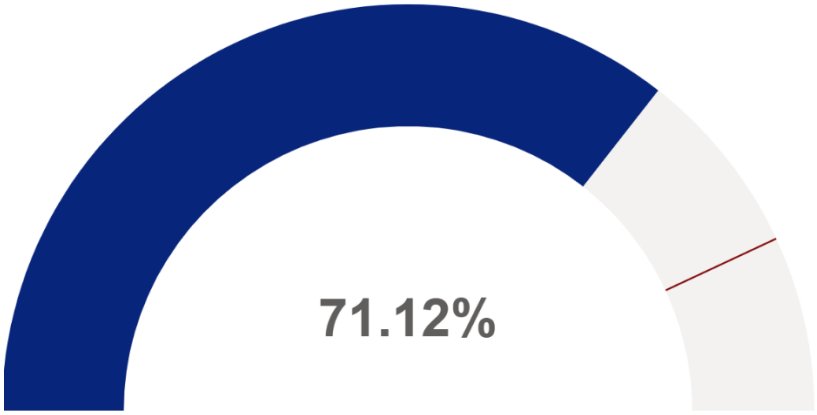
- The Ahmedabad city has the higher level of volume with total number of 19676 orders.
- Ahmedabad is the only City Which Contributes most of the orders.
- The Surat City has the lowest level of Performance with the low order of 17842 orders as compare to other cities.
- The Dairy product Generate the most of the orders almost 38096.
- Food and Beverages are performing the worst nearly fulling 9000 Orders.
- The Dairy product is mostly order in all the cities its high in demand.

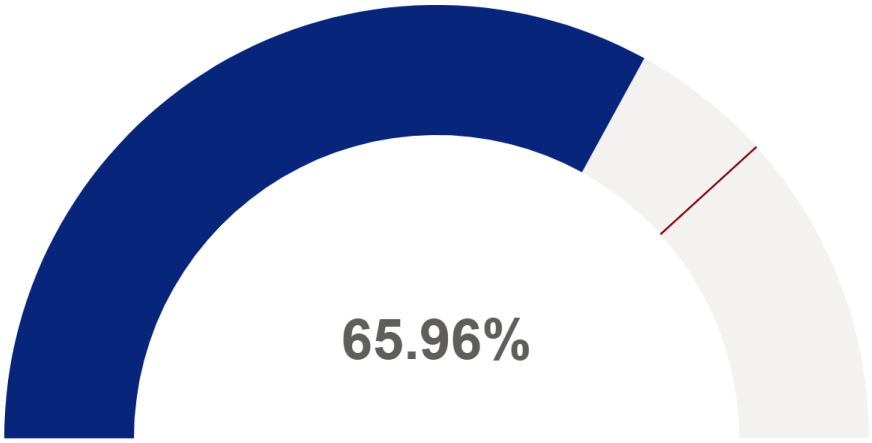
2. Which Stores contribute the highest number of orders?

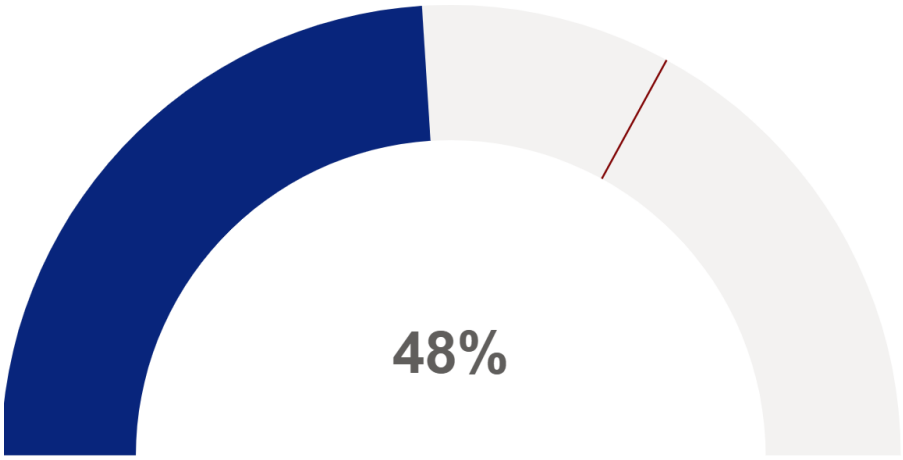


- The Vijay store, Rel fresh, Lotus Mart, Propel Mart, Acclaimed Stores Contribute the highest percentage of order Overall 8.40% to 8.70%.
- Coolblue, Expert Mart, Expression Stores, Elite Mart, Sorefoz Mart, Logic Stores, Atlas Stores, Info Stores, Vivek's Stores Contribute the lowest percentage of order overall 5.85% to 5.64%
- The demand is Concentrated only in few stores which are Contributing most of the orders stores like Vijay Store and Rel Fresh.

3. What is the Order Target at city level?







- The overall on time Delivery performance (OT%) is 71.12% unable to achieved the target which is 86%, We are facing -15% of the delivery gap.
- The overall in full Delivery performance (IF%) is 65.96% unable to achieved the

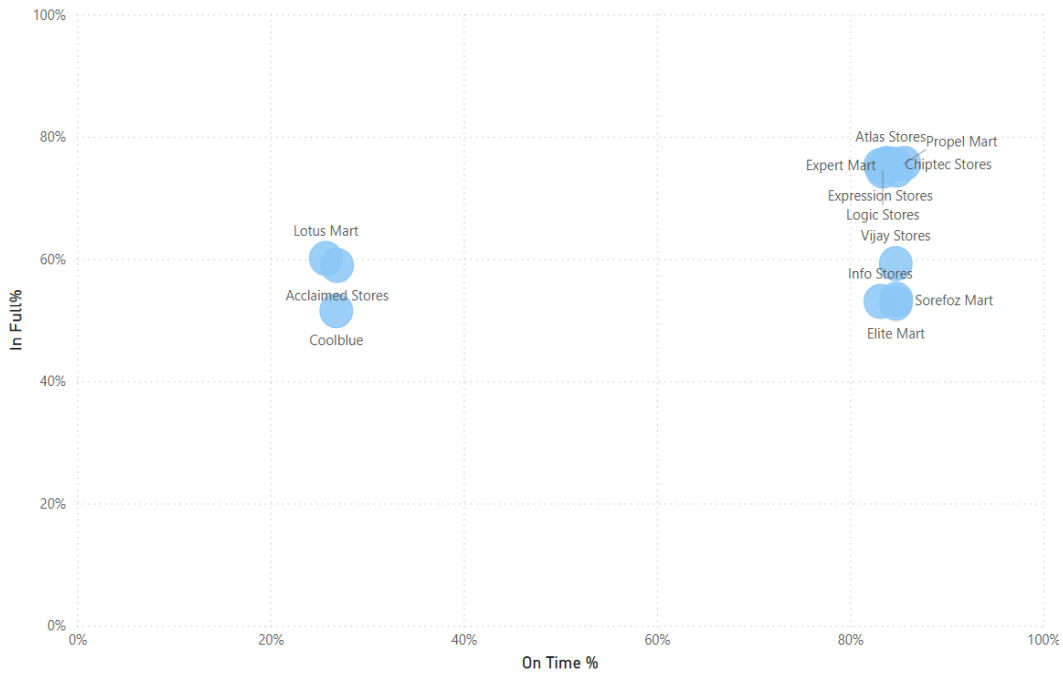
target which is 77%, We are facing -11% of the delivery gap.

- The overall in on time In Full Delivery performance (Onif%) is 48% unable to achieved the target which is 66%, We are facing -18% of the delivery gap.

- Surat is performing better than other cities with 73.67% of on time delivery.
- Vadodara is performing the worst than other cities with 63.69% of in full delivery.
- Surat is performing best in the on time in full deliveries 51% and Vadodara is the lowest performance city overall.
- Almost 30% of the deliveries are late.

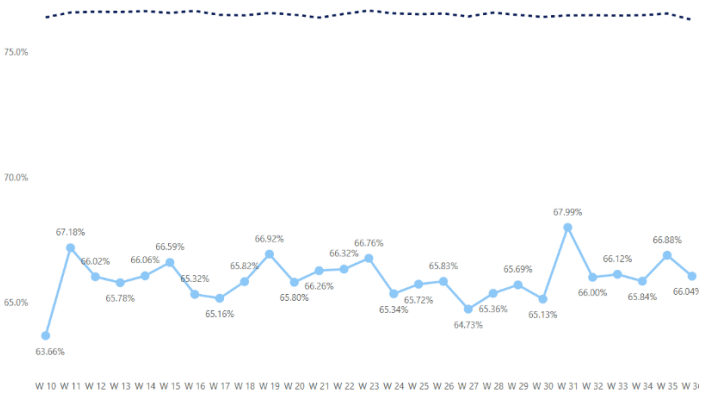
4. What is the service level analysis of the customers?

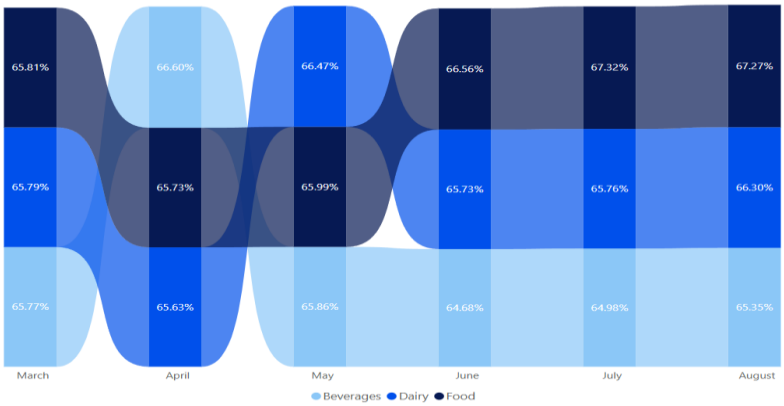
Metric	Acclaimed Stores	Atlas Stores	Chiptec Stores	Coolblue	Elite Mart	Expert Mart	Expression Stores	Info Stores	Logic Stores	Lotus Mart	Propel Mart	Rel Fresh	Sorefoz Mart	Vijay Stores	Viveks Stores
IF%	58.93%	75.48%	75.61%	51.53%	52.74%	75.48%	75.28%	53.05%	74.39%	60.08%	75.62%	74.54%	53.40%	59.23%	75.06%
OT%	26.89%	84.20%	83.76%	26.81%	84.74%	84.62%	83.13%	83.14%	83.39%	25.73%	85.62%	84.84%	84.79%	84.73%	83.42%
OTIF%	15.24%	63.80%	62.97%	13.75%	44.88%	63.61%	62.31%	43.48%	61.96%	16.06%	65.05%	63.04%	45.60%	49.61%	63.04%

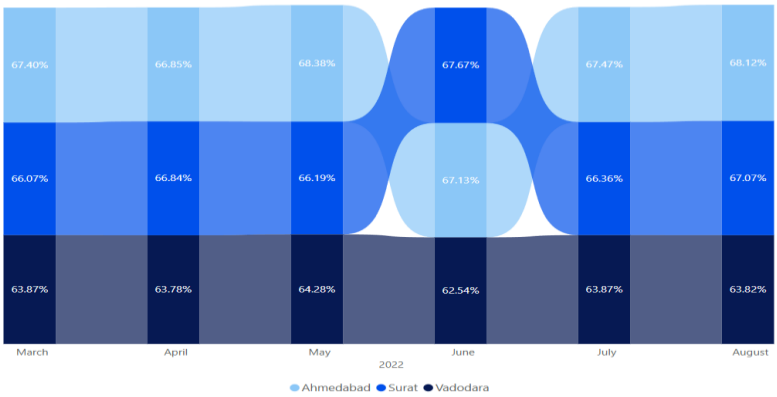


- Propel Mart customer has the highest on time in full performance almost 65 % of the deliveries.
- Locus Mart, Acclaimed Stores, Coolblue are not performing well because there Otif delivery rate percentage is lower than 20%
- Propel Mart customer is the best serve in the supply chain with 85% of the on-time delivery rate.
- Coolblue customer experience the worst service level with 13% of the on time in full delivery.
- Many customers Cluster around 80%-90% OT and 60%-75% IF

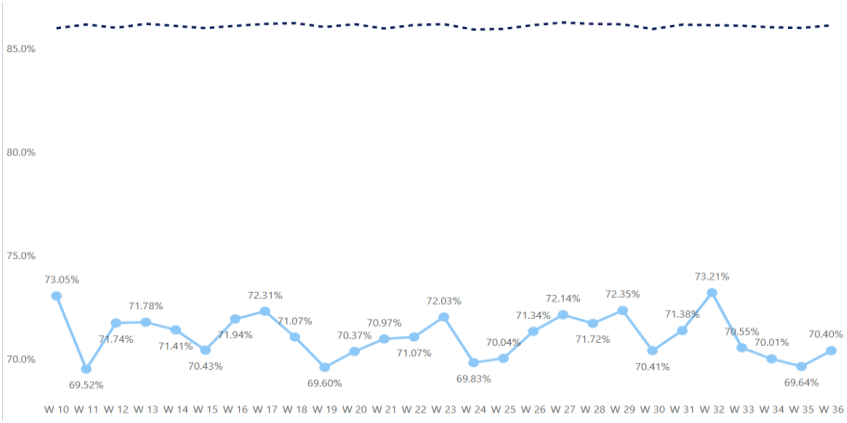
5. What is the Key trend of the Metrics (IF%, LIFR%, OT%, OTIF%, VOFR%)?

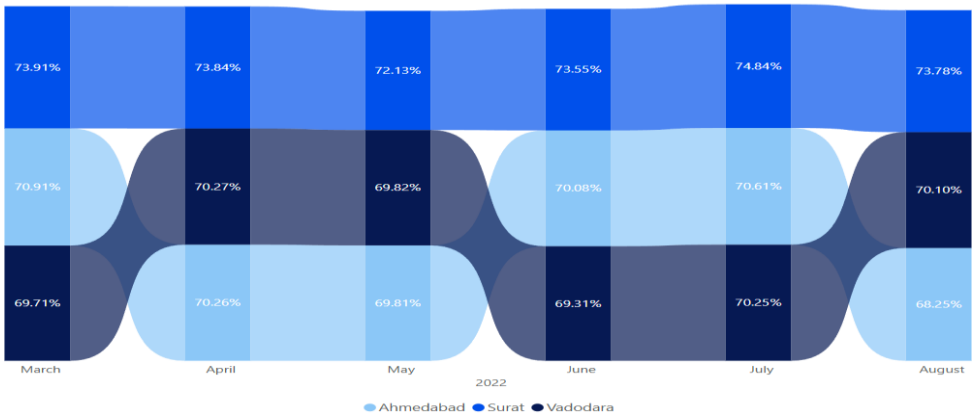


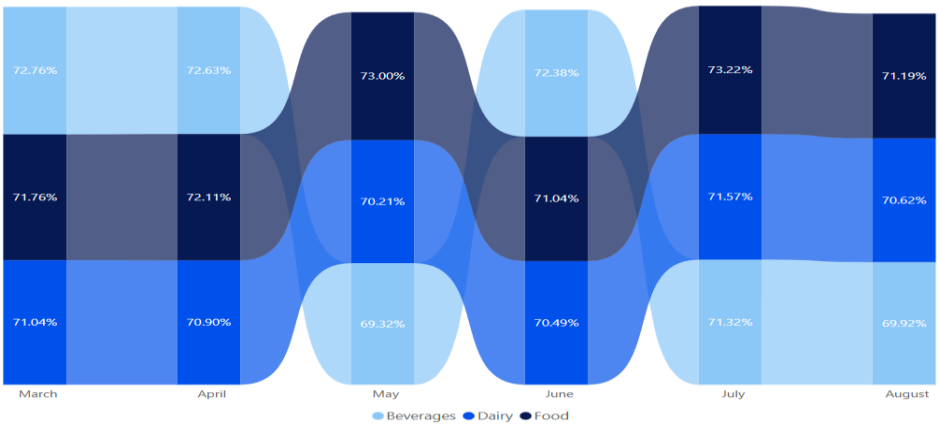




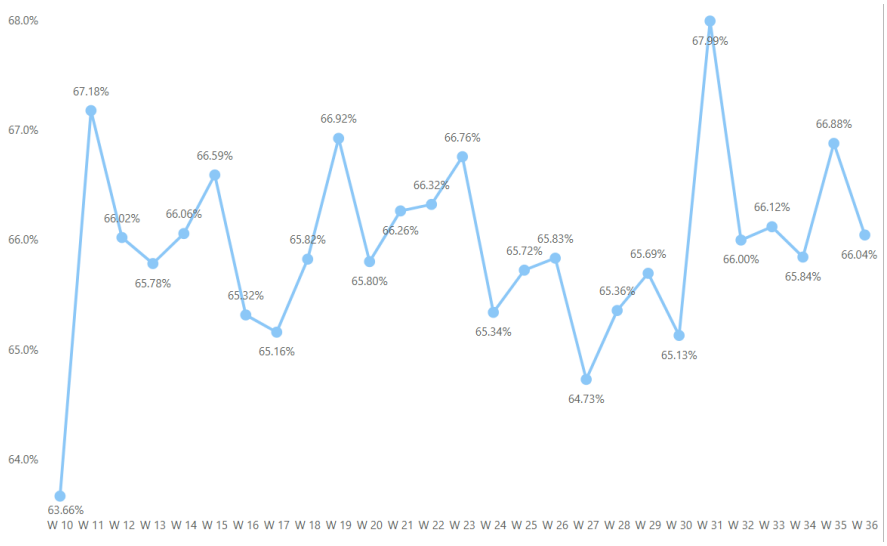
- The weekly IF% fluctuates around 65 to 67 % and the target line in 75 %
- The supply Chain consistently underperforms the target by 8 to 10%
- The IF% Values move between lowest 63% and highest 67-68% performance is stable but the consistency is below target.
- Vadodara consistently performs worst in fulfillment.
- Food Category has the lowest fulfillment rates.

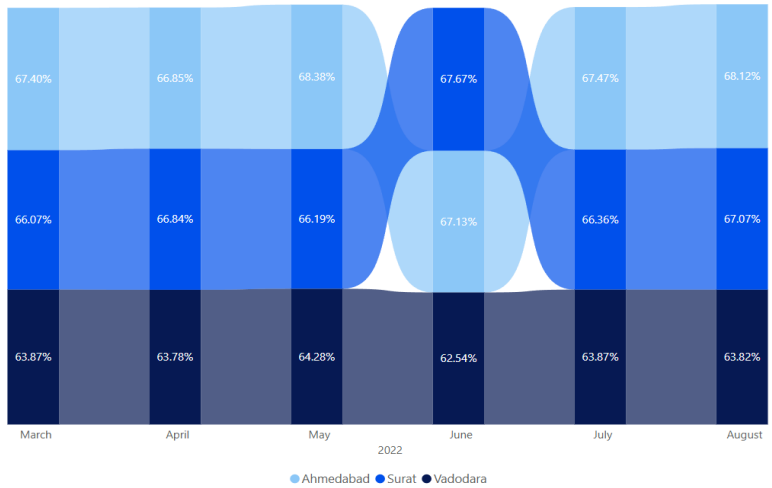


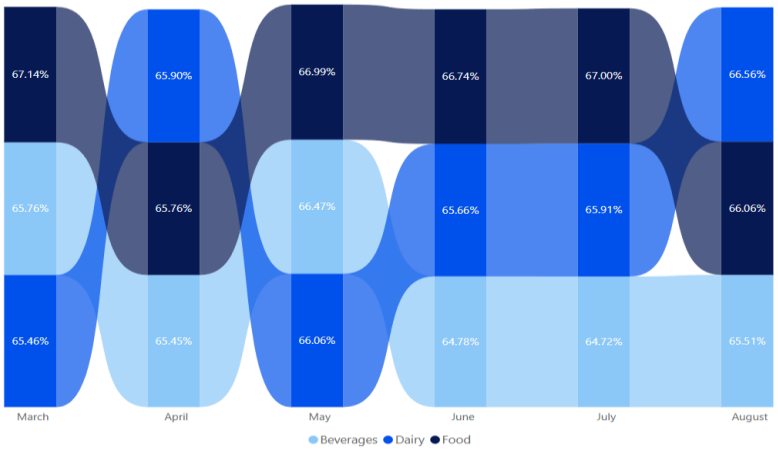




- The weekly OT% fluctuates around 69% to 73% the target is 83%.
- The company is completely missing the OT% target by 7% to 10%.
- The lowest week is week-11 69.5% and week-20 69.6%
- The highest week is week-17 72.3% and week-32 73.2%
- Vadodara has the lowest on time delivery performance 69 to 70%
- Food product has the most delivery delays.

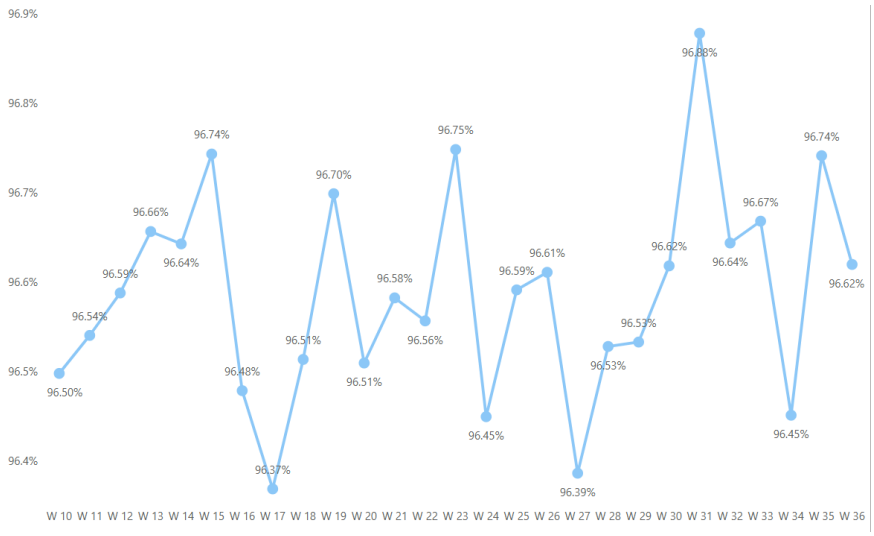


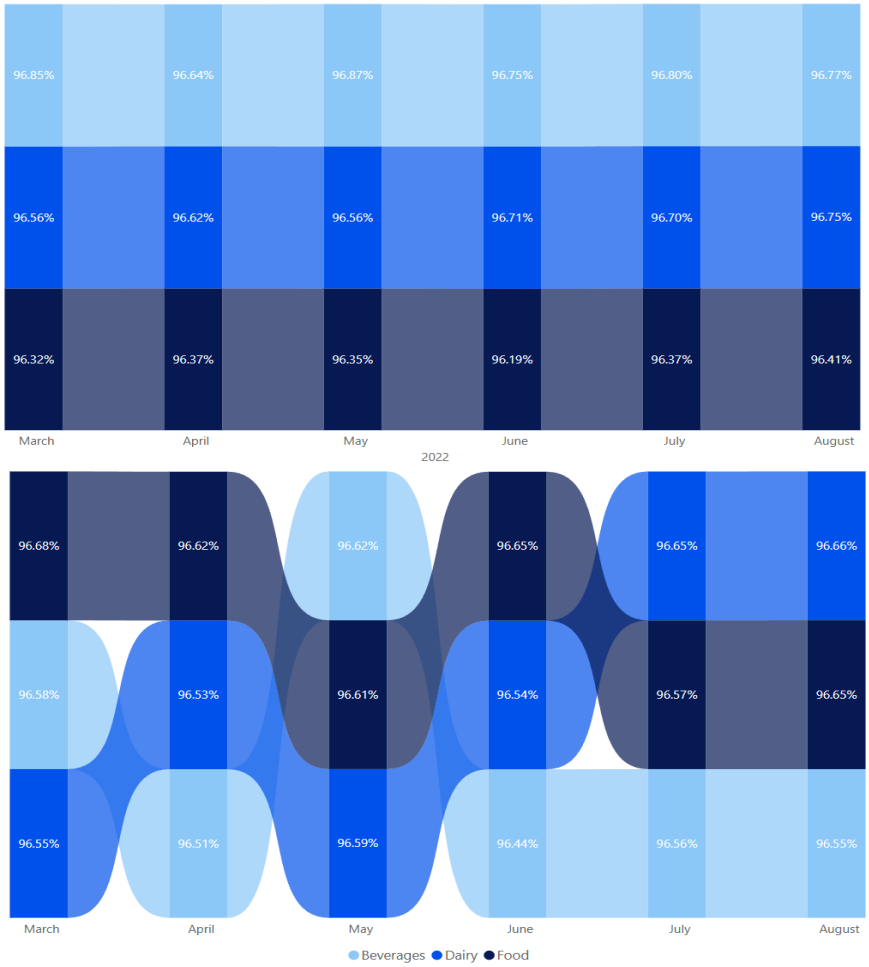




- The weekly LIRF fluctuates around 63% to 67%. Around 33-37% of the items experience shortages but no major improvement.

- Vadodara has the lowest line-item fulfillment performance around 64-65%
- Food products have the lowest fill rate around 64-65%

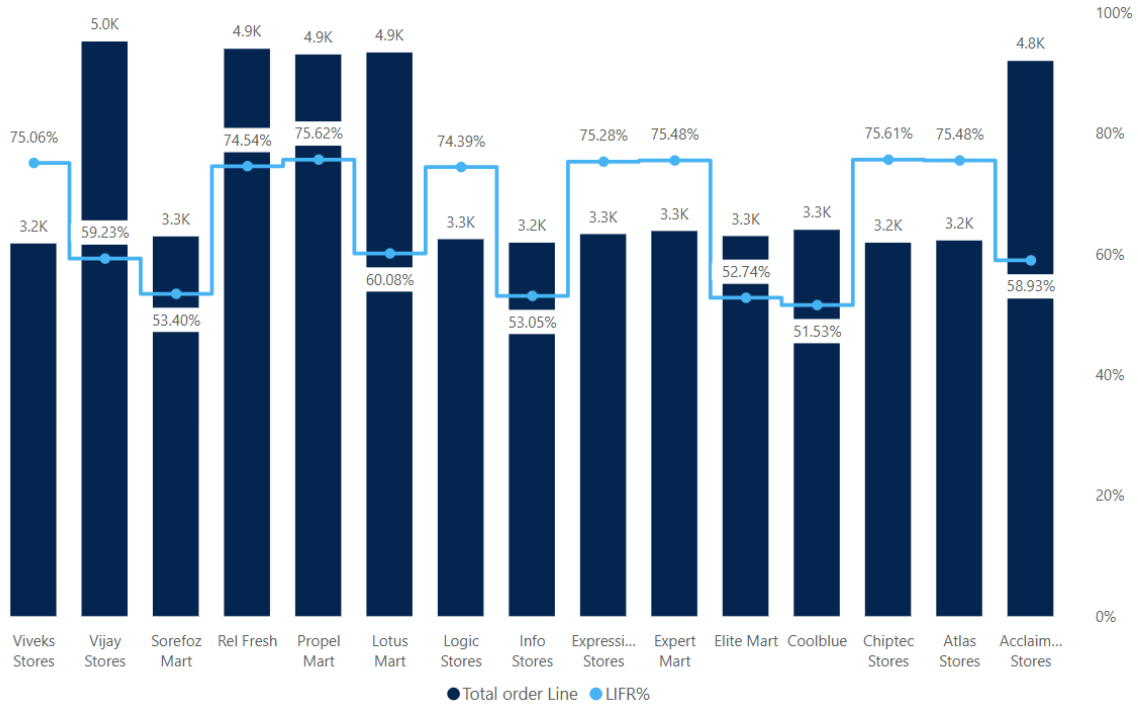




- The Weekly VOFR fluctuated around 96.3% to 96.9% shows very high Fulfillment performance.

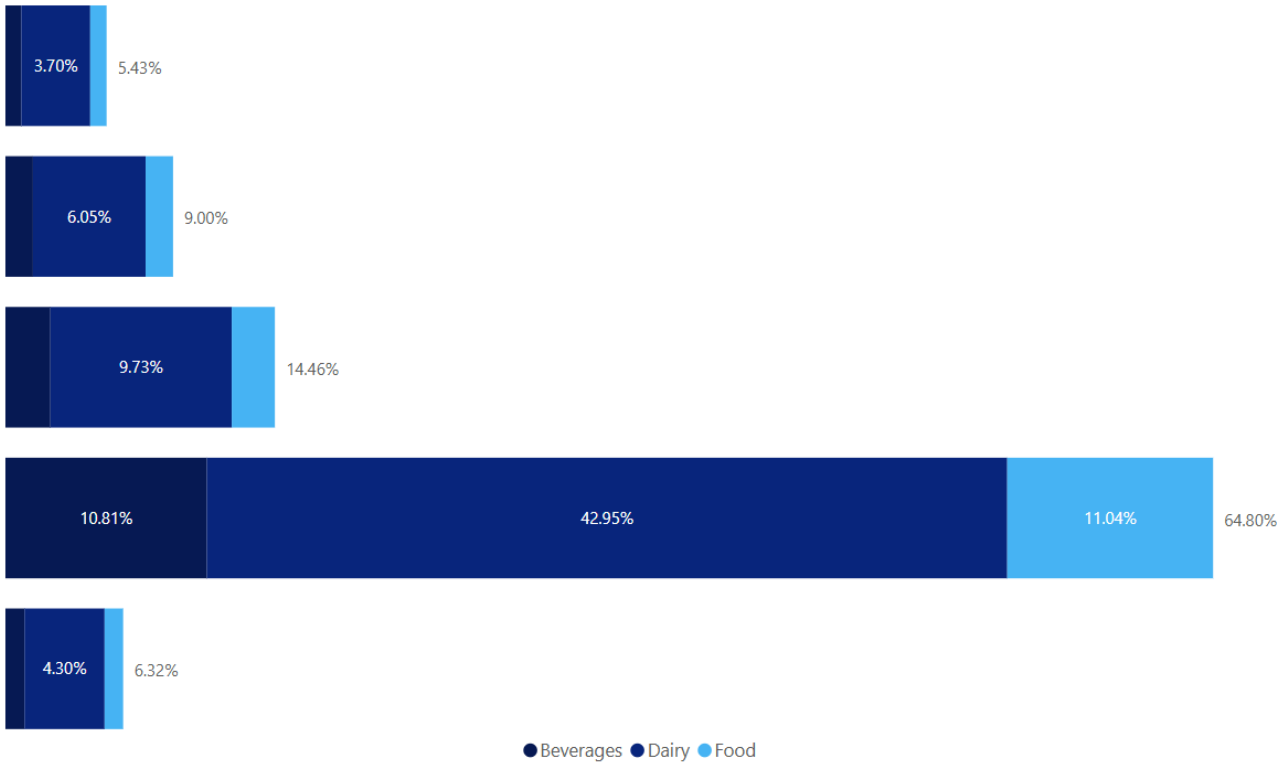
- Vadodara has slightly lower fulfillment performance. (96.4%-96.4%)
- All product has very all high-volume fill rate, food category is slightly lower

6. How is the order line performance?



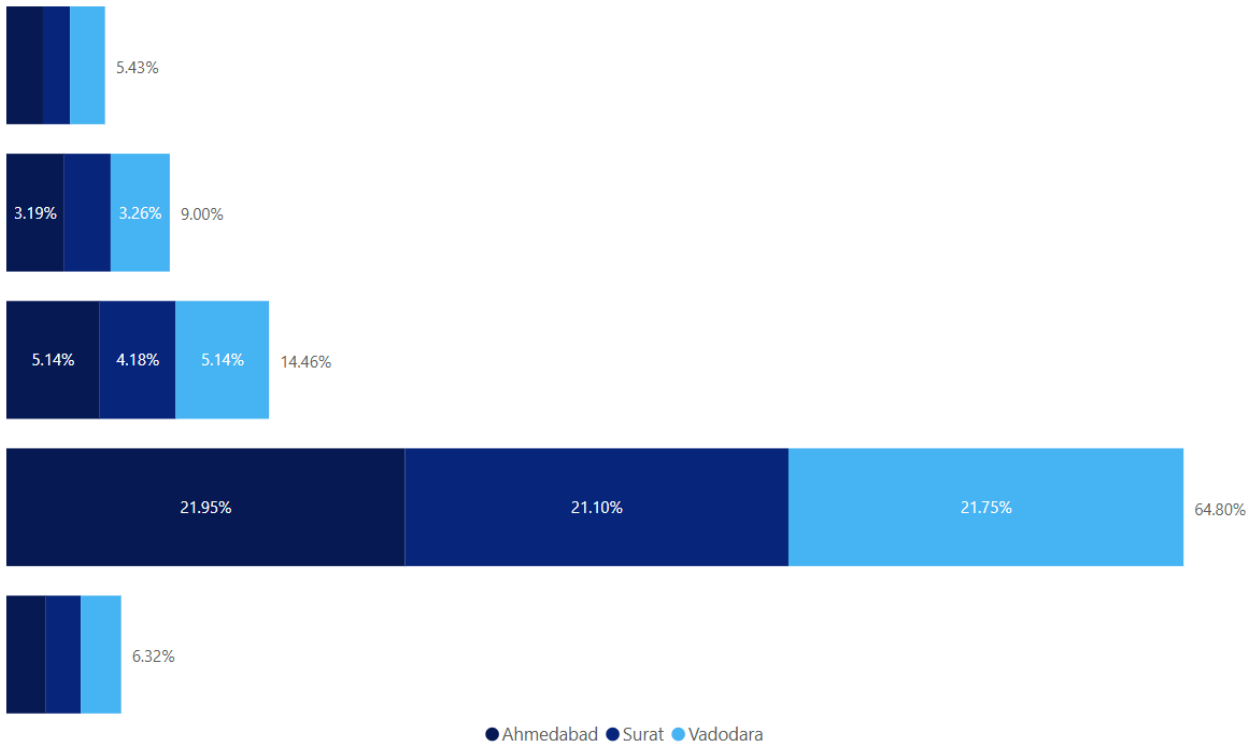
- The company processed 57,000 individual product line orders.
- 1.27M units the total order quality delivered across all orders.
- Only 38k order lines were delivered completely.
- 65.96% of the order line were fulfilled completely and 34% of orders had missing items
- Customers Vijay stores, Rel Fresh, Propel Mart, Acclaimed Stores this are high order volume customers around 4k to 5k orders.
- Customers Elite Mart (52%), Coolblue (51%), Info Stores (53%) with low LIFR% rate.
- Customers with high LIFR% equal to 75% chiptec stores, Atlas Stores, REL Fresh, Propel Mart
- One Third of the order line is incomplete

7. What is the lead line time Analysis?

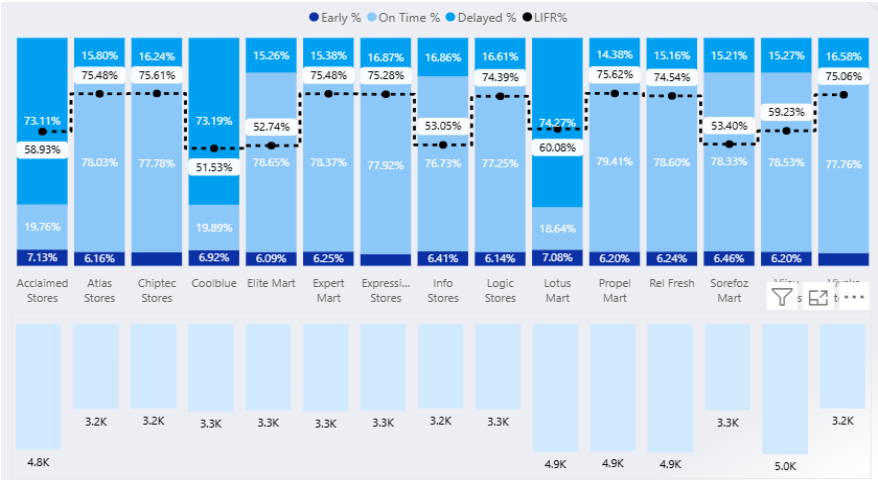


Delayed Days	Beverages	Dairy	Food	Total
-1	584	2454	569	**3607**
0	6171	24521	6306	**36998**
1	1377	5557	1321	**8255**
2	843	3452	841	**5136**
3	486	2112	502	**3100**
Total	**9461**	**38096**	**9539**	**57096**

Delayed Days	Ahmedabad	Surat	Vadodara	Total
-1	1243	1094	1270	**3607**
0	12531	12050	12417	**36998**
1	2934	2389	2932	**8255**
2	1819	1458	1859	**5136**
3	1149	851	1100	**3100**
Total	**19676**	**17842**	**19578**	**57096**



- Around 8255 orders had been one day delayed late
- Approximately 36,998 orders delivered on time
- Around 5,136 orders delayed by 2 days
- Around 3,100 orders delayed by 3 days
- Most on time deliveries come from Dairy product around 42.95% of contribution
- 14.6% deliveries are one day late is the biggest operational issue
- Furthermore ,2 days is 9% and 3 days is 5.43%
- All the cities are performing well with 21% of the delivery rate
- Food and beverages volumes are smaller so their impact is lower



- Most of the customers have 75% to 79% on time deliveries.
- Atlas Stores has Shown 78.03 % of on time delivery which is stable
- Elite Mart has Shown 78.65 % of on time delivery which is stable
- Propel Mart has Shown 79.41% of on time delivery which is stable
- Delayed deliveries are mostly round 14% to 16%
- Acclaimed Stores, Coolblue, Lotus Mart have the highest delayed deliveries around 73% to 74%
- Early deliveries around 6% to 7%
- LIFR% show low for customer Coolblue, Infostore, Sorefoz Mart and Vijay Store, Acclaimed stores around 51% to 59%
- Lotus Mart, Acclaimed Stores, Propel Mart, Rel Fresh, Vijay Stores have the large volume of orders around 4k to 5k.

8. How are the products are performing?

category	product_name	LIFR%	VOFR%	LIFR% by Date	VOFR% by Date
⊟ **Beverages**	AM Tea 100	65.90%	96.63%		
	AM Tea 250	65.34%	96.52%		
	AM Tea 500	65.12%	96.48%		
⊟ **Dairy**	AM Butter 100	66.17%	96.62%		
	AM Butter 250	64.25%	96.38%		
	AM Butter 500	65.64%	96.59%		
	AM Curd 100	66.25%	96.58%		
	AM Curd 250	66.30%	96.61%		
	AM Curd 50	65.57%	96.58%		
	AM Ghee 100	65.96%	96.57%		
	AM Ghee 150	66.46%	96.68%		
	AM Ghee 250	65.50%	96.53%		
	AM Milk 100	66.46%	96.63%		
	AM Milk 250	65.63%	96.62%		
	AM Milk 500	66.85%	96.63%		
⊟ **Food**	AM Biscuits 250	66.41%	96.64%		
	AM Biscuits 500	66.74%	96.62%		
	AM Biscuits 750	66.73%	96.63%		

- All the product has the 96% of the VOFR% delivery is almost in full order quantity.
- The LIFR% is totally low at most the average of 65% means the order line is not fulfilled completely